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RESEARCH ARTICLE

Reproducibility of Pollution–Health Evidence Synthesis Using LLM-Assisted Screening and Extraction

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Abstract

Background: Large language models (LLMs) are increasingly used in evidence synthesis, yet a critical assumption—that identical inputs produce identical outputs—remains largely untested across a complete synthesis pipeline. **Methods:** We quantified the reproducibility of six LLM deployment stacks—each a tuple of (model weights, provider, infrastructure, API layer)—across three local stacks (LLaMA 3 8B, Mistral 7B, Gemma 2 9B served via Ollama on Apple M4) and three API-served stacks (Claude Sonnet 4.5/Anthropic, Gemini 2.5 Pro/Google, GPT-4.1/OpenAI), applied to a two-stage pipeline (abstract screening and data extraction) over a 500-abstract PubMed corpus on fine particulate matter ($PM_{2.5}$) and respiratory health. Each stack was run 10 times under identical configurations (temperature=0, fixed seeds where supported), totalling 35,996 successful calls of 36,000 attempted (99.99% success rate). To probe the deployment-vs.-weights confound, we additionally served `meta-llama/llama-3-8b-instruct` via a cloud endpoint (DeepInfra), holding the weights, prompt source, temperature and seed constant while the rest of the serving stack—including its request-rendering layer—varied. We pre-committed analysis plans for the post-revision experiments to the public Git repository before result computation. The primary outcome was the Exact Match Rate (EMR) with 95% bootstrap confidence intervals. **Results: API-served deployment stacks achieved extraction EMR as low as 0.050 (Claude) despite temperature=0 and fixed seeds, while local stacks achieved EMR=1.000—an approximately 19× gap.** Critically, the same LLaMA 3 8B weights served via a cloud stack (DeepInfra) produced 39% systematic disagreement with the local Ollama stack, showing that identical weights do not yield identical behaviour once the serving stack changes. Across all 100 INCLUDE articles pooled by random-effects meta-analysis, the pooled estimate was robust to non-determinism; in random k=10 subsamples, however, 0.5% of subsamples produced run-dependent reversals of the meta-analytic conclusion. A pairwise-disagreement analysis ($\binom{10}{2}$=45 pairs per item) revealed that 95% of Claude's extraction articles produced at least one differing run pair. BERTScore F1 on metadata saturated near 0.997, masking true narrative variation (Jaccard similarity on rationales ranged 0.38–0.73 for API-served stacks, vs. 0.98 on metadata). **Conclusions:** Non-determinism in API-served deployment stacks poses a quantifiable threat to evidence synthesis reproducibility, particularly for quantitative extraction tasks and small-k literatures. The unit of reproducibility variation is the deployment stack, not the model weights alone. Single-run results should not be treated as definitive; multiple repetitions, provenance tracking, and—where feasible—deterministic local stacks should become standard practice.

1. Introduction

1.1. The Promise of AI in Evidence Synthesis

Evidence synthesis—the systematic process of identifying, appraising, and combining research findings—forms the backbone of evidence-based medicine and public health policy. Systematic reviews and meta-analyses sit atop the hierarchy of evidence, informing clinical guidelines, regulatory decisions, and resource allocation worldwide.[1,2] Yet the process is notoriously labor-intensive: a typical systematic review requires 6–18 months and hundreds of person-hours, with title and abstract screening alone consuming an estimated 25% of total effort.[3]

The emergence of large language models (LLMs) has created unprecedented opportunities to accelerate this process. Recent studies demonstrate that models such as GPT-4,[4] Claude,[5] and Gemini[6] can screen abstracts with sensitivity approaching that of human reviewers,[7,8] extract structured data from published studies with over 90% accuracy,[9,10] and even assist in evidence quality assessment.[11,12] A growing body of work has established LLM-assisted evidence synthesis as a legitimate and rapidly expanding methodological frontier.[13–16] The 2025 Cambridge *Research Synthesis Methods* editorial[17] explicitly invites methodological work characterizing the reproducibility of generative-AI-assisted evidence synthesis, including specific reporting requirements for prompt design, random-seed control, and external validation.

1.2. The Hidden Problem: Non-Determinism

Beneath this promise lies a rarely examined assumption: *if you ask the same question twice, you should get the same answer.* In the context of LLM-assisted evidence synthesis, this means that running the same model with the same prompt, temperature, and random seed on the same abstract should produce identical screening decisions and extraction outputs every time.

This assumption does not hold in practice.

Recent work has demonstrated that commercial LLM APIs exhibit non-deterministic behavior even when configured for maximum reproducibility. Atil et al.[18] showed accuracy variations of up to 15% across repeated identical runs, with best-to-worst performance gaps reaching 70%. Klishevich et al.[19] confirmed that temperature=0 does not guarantee deterministic outputs across any of the major models tested (GPT-4o, Claude 3.5, LLaMA 3.2). At a deeper level, Fu et al.[20] demonstrated that even when surface-level text appears identical, the underlying token probability distributions exhibit substantial instability, suggesting that apparent determinism may mask hidden variation. A philosophical analysis by Karelin et al.[21] traced this non-determinism to three fundamental sources: numerical precision in floating-point operations, architectural choices in distributed inference, and sampling procedures inherent to the generation process.

1.3. Why This Matters for Evidence Synthesis

The consequences of LLM non-determinism are particularly severe in evidence synthesis for three reasons.

First, decisions propagate. A screening decision that flips from “include” to “exclude” in a repeated run does not merely change one data point—it removes an entire study from downstream analysis. In meta-analysis, where pooled effect estimates are sensitive to study composition, a single inclusion change can shift a confidence interval across the null, transforming a statistically significant finding into a non-significant one.[22]

Second, extraction compounds variation. Data extraction involves free-text parsing of complex scientific results—relative risks, confidence intervals, exposure definitions, population characteristics. Even small variations in how an LLM parses “RR=1.05 (95% CI: 1.01–1.09)” can propagate through meta-analytic pooling. Gartlehner et al.[9] reported 95–97%

test-retest reliability in LLM extraction, but this still implies 3–5% of extracted values may change between runs—a rate that, across hundreds of data points, can materially alter pooled estimates.

Third, reproducibility is the foundation of scientific credibility. The broader reproducibility crisis in AI research[23–25] has already eroded trust in computational findings. When LLM-assisted reviews cannot be independently reproduced because the API returns different outputs, the scientific value of those reviews is undermined.[26–28]

1.4. The Gap We Address

Despite the rapid adoption of LLMs in evidence synthesis, **no study has systematically measured how API-level non-determinism propagates through a complete synthesis pipeline**—from abstract screening through data extraction—in a real health domain. This is a critical blind spot: the field is deploying LLMs at scale on the assumption that outputs are stable, without evidence to support that assumption.

Consider the state of current evidence. Jensen et al.[10] report 94.1% inter-session reproducibility for ChatGPT-4o extraction—an encouraging number. But this was measured with only 2 repetitions and evaluated per-field, not per-output. As we demonstrate below, applying a stricter whole-output metric across 10 repetitions yields extraction EMR as low as 5.0%—approximately 19× lower than what per-field metrics suggest. Similarly, Gartlehner et al.[9] report 95–97% test-retest reliability, but with only 2–3 repetitions and no examination of how variation compounds across pipeline stages. Studies measuring non-determinism[18,19] do so on general NLP benchmarks, where the structured, multi-field outputs characteristic of evidence synthesis are absent. And the most influential screening evaluations[7,8,15] report results from a single run, implicitly assuming that a repeat would yield the same result.

We address this gap with the following contributions:

1. Quantifying reproducibility across 10 repeated runs of 6 LLMs (3 local, 3 API) through a two-stage evidence synthesis pipeline;
2. Measuring variation at multiple granularities: whole-output, per-field, per-abstract, and run-pair;
3. **Isolating the deployment effect from the model-size confound** via an additional 10-run experiment in which the same `meta-llama/llama-3-8b-instruct` model is served via a cloud endpoint (DeepInfra) under settings identical to the local Ollama deployment;
4. Testing whether prompt design (variable-length array vs. fixed-slot single-estimate) is the primary driver of cloud-API extraction non-determinism;
5. Cross-validating extraction outputs against two independent reference standards: a within-study majority-vote "silver-internal" and an external silver from DeepSeek-R1 (a reasoning model from a different family);
6. Demonstrating that common "deterministic" API settings (temperature=0, fixed seeds) are insufficient to ensure reproducibility, and that the seed parameter empirically delivers a 6× smaller effect than the deployment paradigm.

2. Related Work

2.1. LLMs in Systematic Reviews

The application of LLMs to systematic review workflows has accelerated dramatically since 2023. Oami et al.[7] evaluated GPT-4 Turbo for citation screening, reporting sensitivity of 0.75 and specificity of 0.99, with prompt tuning improving sensitivity to 0.91. Khraisha et al.[8] extended this to multilingual screening and extraction, finding strong specificity (>0.80) but variable sensitivity across languages and review types.

For data extraction, Gartlehner et al.[9] showed that Claude 2 achieved 96.3% accuracy compared to human reviewers, with test-retest reliability of 95–97%. Jensen et al.[10] found that ChatGPT-4o achieves 92.4% extraction accuracy and 94.1% inter-session reproducibility, explicitly noting residual stochasticity across sessions.

More comprehensive pipelines have also been proposed. The TrialMind system[11] demonstrated that human-AI collaboration improved recall by 71.4% and reduced screening time by 44.2% across 100 systematic reviews. Li et al.[13] described an LLM pipeline for health technology assessment, while Matsui et al.[15] showed that a three-layer GPT-3.5/GPT-4 strategy achieved human-comparable sensitivity. Cao et al.[29] presented otto-SR, a system that automates the entire systematic review lifecycle with LLMs, demonstrating near-human performance across screening, extraction, and synthesis—yet without examining run-to-run reproducibility. Khan et al.[30] proposed a collaborative multi-LLM approach for automated data extraction in living systematic reviews, achieving high accuracy but reporting results from single executions. A recent systematic review of automated meta-analysis[31] found that 57% of studies focus on data processing tasks, yet only 2% have attempted full-process automation—underscoring that evidence synthesis remains a multi-stage pipeline where each step introduces potential variation.

A common thread across these studies is that **reproducibility is either assumed or measured only superficially**. Test-retest reliability, when reported, is based on 2–3 repetitions and does not examine how variation propagates through multi-stage pipelines.

2.2. The Non-Determinism Problem

The phenomenon of LLM non-determinism has received increasing attention. Atil et al.[18] provided the most comprehensive analysis to date, documenting accuracy swings of up to 15% across repeated runs and introducing the TARr@N metric to quantify run-to-run stability. Klishevich et al.[19] confirmed that temperature=0 fails to ensure determinism across GPT-4o, Claude 3.5, and LLaMA 3.2 on code review tasks.

At the token level, Fu et al.[20] showed that apparent text-level agreement masks substantial distributional instability in token probabilities—a finding that suggests surface-level reproducibility metrics may underestimate true variation. Angermeir et al.[32] examined how non-determinism undermines reproducibility in software engineering research, proposing documentation standards to mitigate this.

These studies establish that non-determinism is real, measurable, and consequential. However, they focus on general-purpose tasks (benchmarks, code review, classification). Our study addresses non-determinism specifically in evidence synthesis, where structured, domain-specific outputs amplify the problem.

2.3. Reproducibility in AI Research

The broader AI reproducibility crisis has been documented extensively. Hutson[23] identified unpublished code and sensitivity to training conditions as primary barriers. Haibe-Kains et al.[24] called for structural reform, advocating for mandatory code and model sharing. Kapoor and Narayanan[25] documented data leakage affecting 294 studies across 17 fields, while Han[26] analyzed reproducibility failures in biomedical AI through the lens of model non-determinism.

Recent frameworks have attempted to systematize reproducibility requirements. Desai et al.[27] proposed a taxonomy of reproducibility types, arguing that LLMs require "conceptual replicability" given their proprietary training data. Semmelrock et al.[28] provided a systematic overview of barriers, and Yu[33] argued that computational reproducibility alone is insufficient for trustworthy AI.

2.4. Provenance and Transparency

Provenance tracking—recording the lineage of data, computations, and decisions—has been identified as a key enabler of AI reproducibility. Longpre et al.[34] audited 1,800+ datasets, finding license omission rates above 70%. Liang et al.[35] analyzed 32,111 model cards on Hugging Face, finding that documentation of limitations and environmental impact remains sparse. The Foundation Model Transparency Index[36] scored 14 major developers across 100 indicators, finding average transparency of only 58/100.

Technical solutions for provenance include the Workflow Run RO-Crate[37] and the W3C Provenance (PROV) Data Model—a W3C Recommendation defining an interoperable standard for representing the entities, activities, and agents involved in producing a piece of data.[38] Longpre et al.[39] argued at ICML 2024 that current practices for data authenticity and provenance are "fundamentally broken," calling for comprehensive reform.

Our provenance approach builds on the lightweight protocol proposed by Rover,[22] using SHA-256 hashing of inputs, outputs, and configurations to create an auditable trail without requiring infrastructure beyond standard file storage.

2.5. Positioning: What Existing Studies Do Not Address

Table 1 synthesizes the landscape of prior work and highlights the methodological gaps that motivate our study. Three critical gaps emerge. **First**, studies that evaluate LLMs for evidence synthesis[7,9,10,15] measure accuracy against human judgments but do not measure self-consistency across repeated runs—or do so with only 2–3 repetitions and per-field metrics that can mask whole-output variation. **Second**, studies that measure LLM non-determinism[18–20] do so on general-purpose benchmarks (classification, code review), not on the structured, domain-specific tasks characteristic of evidence synthesis. **Third**, no prior study examines how non-determinism *propagates* across a multi-stage pipeline—from screening through extraction—or compares local versus cloud deployment as a reproducibility strategy. Our study addresses all three gaps simultaneously.

Table 1. *Comparison of our study with prior work. "Runs" indicates the number of repeated executions under identical conditions. "Full pipeline" indicates whether both screening and extraction were evaluated in the same study. Shading highlights our study's distinguishing features..*

Study	Domain	Task	Runs	Metric	Local	Cloud	Pipe.
Oami (2024)	Health	Screening	1	Accuracy	—	✓	—
Oami (2025)	Health	Screening	1	Acc.+cost	✓	✓	—
Khraisha (2024)	Health	Scr.+Extr.	1	Accuracy	—	✓	—
Gartlehner (2024)	Health	Extraction	2–3	Field acc.	—	✓	—
Jensen (2025)	Health	Extraction	2	Field acc.	—	✓	—
Matsui (2024)	Health	Screening	1	Accuracy	—	✓	—
Atil (2024)	General	NLP bench.	10+	Acc. var.	—	✓	—
Klishevich (2025)	Softw.	Code review	5+	Determinism	✓	✓	—
Fu (2026)	General	Token dist.	—	Distrib.	—	✓	—
Li (2025)	Multi	Meta-anal.	—	Survey	—	—	—
Cao (2025)	Health	Full SR	1	Accuracy	—	✓	✓
Khan (2025)	Health	Extraction	1	Field acc.	—	✓	—
This study	**Health**	**Scr.+Extr.**	**10**	**EMR+CI**	✓	✓	✓

3. Methods

3.1. Study Design

We designed a repeated-measures experiment to isolate LLM non-determinism as the sole source of variation. We held the corpus, prompts, model configurations, and random seeds constant across all runs. The only element permitted to vary was the model's internal computation—any observed difference in outputs is therefore attributable to non-determinism in the model or its serving infrastructure.

Figure 1 provides an overview of the experimental pipeline.

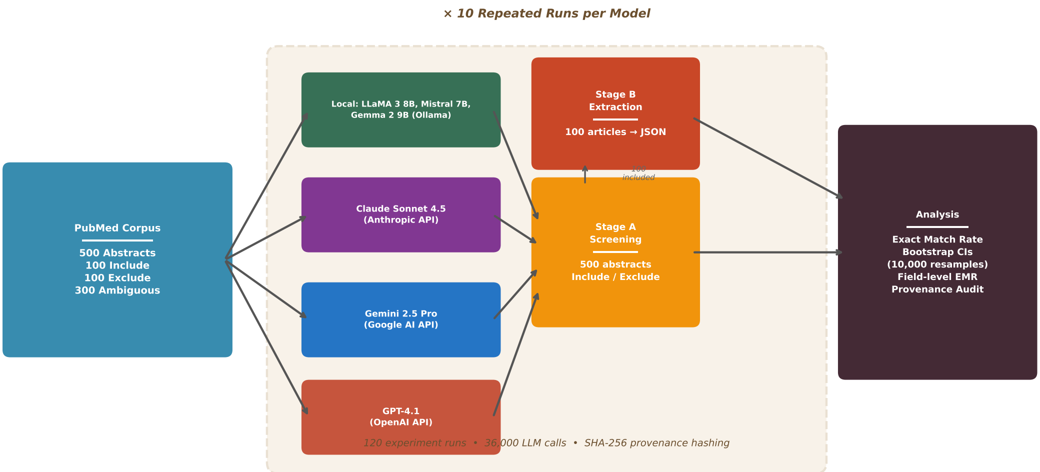

Figure 1. *Experimental pipeline. A 500-abstract PubMed corpus on* $PM_{2.5}$ *and respiratory health is processed by six deployment stacks: three local Ollama/Apple M4 stacks (LLaMA 3 8B, Mistral 7B, Gemma 2 9B) and three API-served stacks (Claude Sonnet 4.5/Anthropic, Gemini 2.5 Pro/Google, GPT-4.1/OpenAI). Each stack is run 10 times under identical configurations (temperature=0, fixed seed where supported). Stage A (screening) produces binary include/exclude decisions over all 500 abstracts; Stage B (extraction) produces structured JSON over the 100 INCLUDE articles. All outputs are SHA-256-hashed for provenance tracking. Reproducibility is quantified via the Exact Match Rate (EMR) with 95% percentile bootstrap confidence intervals..*

3.2. Unit of Analysis: Deployment Stacks

We treat the unit of analysis as the *deployment stack*—the tuple (*model_weights*, *provider*, *infrastructure*, *API_layer*)—rather than the model weights alone. For the three local stacks, every component of the tuple is transparent and pinned: open-weight checkpoints with verifiable hashes (LLaMA 3 8B, Mistral 7B, Gemma 2 9B), served by Ollama v0.15.5 on a single Apple M4 chip (24 GB RAM, single replica, single execution thread), through Ollama's local HTTP endpoint. For the three API-served stacks, only the provider and the client-visible API surface are observable; the underlying weight checkpoint version, GPU topology, batching policy, and intermediate-precision routines are opaque and may change without notice. This framing matters because the local-vs-cloud comparison constitutes a *partial quasi-isolation probe*: when two stacks share approximately matched weights (e.g., `meta-llama/llama-3-8b-instruct` served locally via Ollama and via DeepInfra) but differ in infrastructure and serving layer, any divergence in their outputs is attributable to the non-weight components of the tuple. Throughout this paper we use "deployment stack" for the experimental unit and reserve "model" for the weights alone.

3.3. Corpus Construction

We constructed a 500-abstract corpus from PubMed covering the association between $PM_{2.5}$ and respiratory health outcomes (Table 2). We chose this domain because it is one of the most extensively studied areas in environmental epidemiology, with Yu et al.[40] estimating approximately 1 million premature deaths annually from short-term $PM_{2.5}$ exposure. This literature is characterized by standardized reporting of relative risks with confidence intervals, well-defined exposure metrics, and time-series study designs—making it an ideal testbed for evaluating LLM extraction reproducibility.[41,42] Prior work on particulate matter and respiratory morbidity has shown that machine learning models can capture the complex, lagged relationships between PM_{10} concentration and hospital admissions,[43] further motivating a domain where both the epidemiological patterns and the computational methods are well established.

Table 2. *Corpus composition for the $PM_{2.5}$ and respiratory health abstract screening task..*

Category	Count	Purpose
Clearly include	100	Positive controls (meet all inclusion criteria)
Clearly exclude	100	Negative controls (clear exclusion reasons)
Ambiguous	300	Primary test set (borderline cases)
Total	**500**	

Inclusion criteria required: (1) original epidemiological study, (2) $PM_{2.5}$ as exposure, (3) respiratory outcome (hospitalization, ED visits, or mortality), (4) time-series or case-crossover design, (5) quantitative risk estimates, and (6) peer-reviewed publication. Articles were sourced from PubMed using the E-utilities API with a broad query yielding 573 candidate records, supplemented by 475 exclusion candidates and a design-filtered subset of 222 records. The final corpus spans 148 journals and publication years 1994–2026, with a mean abstract length of 1,873 characters.

Gold standard construction. The 200 clear-category abstracts (100 include + 100 exclude) were labeled by an automated classification rule based on inclusion-criteria keyword matching, validated against a held-out 50-abstract validation set. No errors were observed on that set, which by the rule of three bounds the true error rate at ≈6% with 95% confidence and no tighter; we therefore report that bound rather than describing the rule as 100% precise, since zero errors in 50 trials cannot establish a zero error rate. The 300 ambiguous abstracts received heuristic labels from the same rule operating in low-confidence regions; these labels are reported separately in all accuracy computations and are explicitly flagged as not human-validated. We do not retroactively present these heuristic labels as a human gold standard.

In response to peer review, we additionally conducted a **dual-independent human-labeling validation** on a stratified subset of 100 abstracts (25 clear-include + 25 clear-exclude + 50 ambiguous, stratified by heuristic category × year-tertile, seed=42). The labeling protocol, screening templates, and agreement script were pre-registered on the Open Science Framework *before* any label was collected (https://doi.org/10.17605/OSF.IO/FGN3E, frozen 2026-05-12; the two label sets were returned on 2026-07-15 and 2026-07-29). Two reviewers, each blinded to the other's decisions, independently classified every abstract via the protocol in Supplementary §4; we computed Cohen's κ on the 3-class outcome (include/exclude/uncertain) and on the binary include-vs.-exclude collapse. Agreement fell **below the pre-specified Cochrane target** of $\kappa \geq 0.80$: $\kappa = 0.529$ (SE=0.074, 95% CI [0.383, 0.674]) on the 3-class outcome and $\kappa = 0.556$ (95% CI [0.400, 0.712]) on the binary collapse, with 75% raw agreement and 25 discordant abstracts (confusion matrix and full agreement statistics in Supplementary §4). The shortfall against the target is itself statistically decisive ($z = -3.65$, $p = 1.3 \times 10^{-4}$), whereas $n = 100$ is not sufficient to place κ within a single descriptive band—the interval spans "fair" through "substantial"—so we report the value against the threshold rather than assigning it a label. Two objections can be foreclosed immediately. The low coefficient is not an artefact of skewed marginals: the prevalence index is 0.189 and PABAK (0.558 on the binary collapse) is nearly identical to κ, so the "kappa paradox" is not operating here. Nor is it an artefact of the three-class scale: linearly and quadratically weighted κ (0.541 and 0.548, computed on the ordinal ordering include–uncertain–exclude) move the estimate by less than 0.02. The pre-registration specified in advance what to do in this event—report the value transparently, examine the disagreement pattern qualitatively, amend the protocol to address the ambiguity, and file a sub-registration update—and we followed that sequence rather than revising the target. The disagreement is not diffuse but directional, and this is testable: the discordant cells of the binary table are 2 versus 19 (exact McNemar $p = 2.2 \times 10^{-4}$), and marginal homogeneity is rejected on the full 3-class table (Stuart–Maxwell $\chi^2(2) = 15.4$, $p = 4.6 \times 10^{-4}$), with a bias index of 0.179. One rater is systematically more inclusive than the other, which is the signature of divergent protocol reading rather than of random rater noise. The source is specific: 17 of those 19 asymmetric discordances turn on inclusion criterion 5 (a quantitative effect estimate), whose v1.1 wording supported two defensible readings—requiring the numeric estimate and interval to be present, or accepting an explicit statement that the study reported them. The v1.1 decision table compounded this by specifying outcomes for failure in two or more criteria and for a borderline single criterion, but not for clear failure in exactly one, which is precisely the configuration of those 17 abstracts. Neither rater departed from the protocol; the protocol was underspecified. Protocol v1.2 therefore splits criterion 5 into three levels: values present (criterion satisfied), the effect stated but its values absent (*uncertain*, since abstract-only screening cannot verify what the abstract omits), and no estimate or mention at all (*exclude*, since total absence is reasonable evidence that no extractable estimate exists). The amendment also separates structural criteria (original study, $PM_{2.5}$ exposure, respiratory hospitalization, English), where a single clear failure excludes, from conditional ones (design, effect reporting), where a single failure yields *uncertain* unless the information is absent rather than merely unreported. A blinded recalibration round restricted to the 25 discordant items is underway. We stress what that round can and cannot establish. Because re-measurement is conditioned on prior disagreement, the resulting figure is *not* a second estimate of κ and we do not report it as one: recomputing a whole-corpus coefficient after selectively re-rating only the discordant items would drive the value upward by construction, and would cross the Cochrane threshold at a resolution rate that carries no evidential content. We therefore report it as post-hoc reconciliation agreement conditional on the initially discordant items, and $\kappa = 0.529$ remains the study's agreement estimate. Establishing that the v1.2 amendment actually repairs the ambiguity would require a fresh independent sample rather than a re-rating of the same items, which we identify as the appropriate next step rather than claim as a result here. Residual discordances go to a consensus meeting, and items without consensus are settled by the senior author (Y.d.S.T.) as tie-breaker, exactly as pre-registered. Assigning that role to the senior author rather than to the first author is deliberate: the first author developed the pipeline whose outputs are scored against this gold standard, so placing final adjudication with a co-author who did not build it keeps the reference standard independent of the system under evaluation. Tie-breaks apply only to residual discordant items, are adjudicated against the written criteria without access to any model output, and are logged with the criterion invoked, so each adjudicated item remains auditable.

Stratifying agreement exposes something the pooled coefficient hides, and it bears directly on how our accuracy numbers should be read. On the 25 clear-*exclude* abstracts the two raters agreed unanimously (25/25, PABAK=1.00; κ is undefined there because expected agreement equals one—a degenerate-marginals case that is precisely why we report PABAK alongside). On the 25 clear-*include* abstracts agreement was 68% (κ=0.359), and on the 50 ambiguous abstracts 66% (κ=0.398). The automated rule's include side fares worst under human scrutiny: of the 25 abstracts it called clearly includable, one rater endorsed inclusion for 13 and the other for 21, and 8 of the 25 discordances fall inside that stratum. The gold standard is therefore **asymmetrically valid**. Specificity rests on the unanimous exclude side and is trustworthy; sensitivity rests on include-side labels that the human raters substantially contest, and is correspondingly attenuated. We report this rather than pool it away, and we read all sensitivity figures in this paper as lower bounds.

For extraction, we constructed two independent reference standards: (a) a *silver-internal* consensus from majority vote across all 6 models × 10 runs (60 votes per item), and (b) a *silver-external* consensus from DeepSeek-R1 (5 runs × 100 items, majority vote per field). DeepSeek-R1 was selected for the silver-external role on three independence axes: (i) it is served by a provider outside the four tested in our main analysis, (ii) it was trained predominantly on a different (Chinese-led, mixed open/closed) corpus, and (iii) it uses an explicit-reasoning decoding paradigm distinct from the standard chat models evaluated here, which together make correlated systematic errors with the evaluated stacks unlikely. Cross-validation between these two silvers yields Spearman ρ=0.845 on `effect_estimate` and 80% agreement across numeric fields (consensus subset, see Supplementary §14), supporting their use as comparative anchors. A dual-human extraction reference is being built for the INCLUDE items of the human gold standard, and its provenance requires a disclosure. The extraction subset used in earlier drafts of this work was drawn from the silver-internal standard *before* any human label existed; of those 25 items only 13 are included by first-round human agreement, and 15 abstracts the raters agree on were absent from it altogether. Scoring the models against a reference derived from model output would be circular, so the extraction subset has been rebuilt from the human gold standard instead. The overlap is reported as a result in its own right—direct evidence that LLM-based screening admits material that human raters reject—and its final value is fixed once discordance resolution closes the gold standard (**[PENDING: final overlap, rebuilt subset size, and per-field extraction agreement]**).

Because our primary metric (EMR) measures self-consistency rather than accuracy, gold-standard quality affects only the secondary accuracy analysis; we report secondary accuracy against three sources (heuristic, silver-internal, silver-external—the latter computed on the consensus subset only, see Supplementary §14; plus the dual-human gold standard for screening) to triangulate the result. Given the asymmetric validity documented above, we treat the exclude side of every accuracy comparison as firm and the include side as a lower bound.

3.4. Deployment Stacks

We selected six deployment stacks spanning local and API-served paradigms (Table 3). Each stack instantiates the tuple (*model_weights*, *provider*, *infrastructure*, *API_layer*) defined in Section 3.2. The three local stacks cover three open-weight families served on a single device; the three API-served stacks cover three proprietary providers accessed over the public network. The three local stacks pair open-weight checkpoints—LLaMA 3 8B,[44] Mistral 7B,[45] and Gemma 2 9B[46]—with Ollama v0.15.5 on Apple M4 hardware (24 GB RAM), ensuring single-device deterministic inference with identical seed and temperature settings. The three

Table 3. *Deployment-stack configurations. The four-component tuple (weights, provider, infrastructure, API_layer) is shown across the first four columns. All local stacks were served via Ollama with deterministic settings. Claude does not expose a seed parameter; Gemini accepts seeds but does not guarantee determinism; GPT-4.1 accepts seeds via the API. Local stacks are 7–9B-parameter open-weight checkpoints; API-served weights are proprietary and approximately >100B parameters, so the local-vs-API comparison varies multiple tuple components simultaneously (see Section 4.8 for a single-component probe)..*

Weights	Provider	Infrastructure	API layer	Temp. / Seed	top_p	Max tokens
LLaMA 3 8B	Meta (open)	Ollama v0.15.5 / Apple M4 / single replica	local HTTP /api/generate	0 / 42	1.0	2,048
Mistral 7B	Mistral (open)	Ollama v0.15.5 / Apple M4 / single replica	local HTTP /api/generate	0 / 42	1.0	2,048
Gemma 2 9B	Google (open)	Ollama v0.15.5 / Apple M4 / single replica	local HTTP /api/generate	0 / 42	1.0	2,048
Claude Sonnet 4.5 (20250929)	Anthropic	opaque (production)	Messages API v1	0 / —	1.0	2,048
Gemini 2.5 Pro	Google	opaque (production)	generateContent v1	0 / 42	1.0	8,192†
GPT-4.1	OpenAI	opaque (production)	Chat Completions v1	0 / 42	1.0	2,048

†Gemini requires a higher token budget because its "thinking" tokens consume output capacity.

API-served stacks accessed proprietary weights through their respective providers: Claude Sonnet 4.5[47] via the Anthropic Messages API (no seed parameter available), Gemini 2.5 Pro[6] via the Google AI `generateContent` API (seed set to 42), and GPT-4.1[48] via the OpenAI Chat Completions API (seed set to 42). All stacks were queried via HTTP (urllib-based, no SDK dependencies) to ensure uniform request handling. API calls included up to 2 retries with exponential backoff on timeout or rate-limit errors; retried calls may route to different server nodes, introducing a minor additional source of variation for the API-served stacks.

3.5. Sources of Non-Determinism in Distributed Inference

It is useful to distinguish two classes of factors that can plausibly contribute to LLM non-determinism across deployment stacks. *Cloud-deployment factors* are those intrinsic to running a model on hardware accessed over a network: end-to-end network latency, request routing across geographically distributed endpoints, shared physical hardware, and multi-tenancy isolation policies. *Production serving-infrastructure factors* are specific design choices in the LLM inference stack: tensor parallelism with non-deterministic all-reduce, FlashAttention kernel non-determinism, dynamic continuous batching that varies request composition between calls, speculative decoding fast-paths, mixed-precision (BF16/FP16) accumulation, and prefix-caching state shared across requests. These two classes are conceptually distinct: a request that traverses the network can still be served on a deterministic infrastructure, and vice versa, an infrastructure with the non-deterministic features above can be exercised under local-network conditions. Our desconfound experiment (Section 4.8) helps separate them empirically by holding the weights constant while varying the rest of the stack tuple.

Table 4 maps six of these mechanisms onto each of the six deployment stacks studied here, classifying each cell as documented (✓), *likely* (consistent with public information from the provider's serving infrastructure), *possible* (compatible with public information but not directly confirmed), or precluded by stack design (—). Attribution is necessarily *inferential* for the API-served stacks because the production serving stacks of OpenAI, Anthropic, and Google are closed-source; the table reflects what is publicly documented or strongly implied by published architecture descriptions, not what we directly measured per request.

Table 4. *Mapping of candidate non-determinism mechanisms onto the deployment stacks evaluated in this study. ✓ = documented in public materials (e.g., provider engineering blog, vendor docs, or directly observable from API behaviour);* likely *= strongly consistent with public information about the stack but not directly attested per request;* possible *= compatible with public information but unconfirmed; — = precluded by the stack's design (e.g., single-replica local inference cannot use multi-GPU all-reduce). Attribution for the three API-served stacks is inferential; for the three local Ollama/M4 stacks, the absence of every listed mechanism follows directly from the single-device, single-execution-thread serving configuration. The two Together-AI / DeepInfra cloud columns refer to the same-weights desconfound stack (Section 4.8)..*

Mechanism	Local (Ollama/M4)	Cloud (DeepInfra/L3-8B)	API-Claude	API-Gemini	API-GPT-4.1
1. Floating-point non-associativity (BF16/FP16 mixed-precision accumulation)	—	✓	likely	likely	likely
2. Tensor parallelism / multi-GPU all-reduce reduction order	—	likely	likely	likely	likely
3. FlashAttention / fused-kernel non-determinism	—	likely	likely	likely	likely
4. Dynamic continuous batching (batch composition varies between calls)	—	likely	✓	✓	✓
5. Speculative decoding / draft-model fast-paths	—	possible	likely	likely	likely
6. Prefix-caching state shared across requests	—	possible	possible	possible	possible

The mapping is used in two ways in our discussion. First, the residual disagreement observed in the same-weights cloud experiment (Section 4.8) is most parsimoniously attributable to mechanisms 1–4, because mechanisms 5–6 are not standard for an 8B-parameter model under modest load. Second, the larger gap observed for the three API-served stacks of much-larger proprietary weights is consistent with all six mechanisms being active simultaneously; we therefore do not attempt single-mechanism attribution for those stacks in the present study, and we mark this as a key direction for future work (Section 5.6).

3.6. Prompt Development

Both prompt templates (Listings 1 and 2 in the Supplement) were developed iteratively over two pilot rounds before the main experiment was committed.

Round 1. An initial draft was tested on 10 randomly drawn abstracts from the corpus (5 include + 5 exclude) using LLaMA 3 8B locally. Output JSON validated against the schema for 9 of 10 calls; one failure was due to embedded Markdown code-fences. The failure case motivated the explicit instruction "Respond with a JSON object only. Do not include any text outside the JSON." in both prompts.

Round 2. The revised prompt was tested on the same 10 abstracts plus 10 additional abstracts drawn from the ambiguous stratum. Output validated for 20/20 calls. We additionally manually inspected the rationale text for coherence (no contradictions between rationale and decision); none were identified.

The final prompts were committed to the public repository on 2026-02-11 (commit `75292f2`) and held constant across all 6 models and all 120 experimental runs. We did not adapt prompts to individual model capabilities beyond the generic JSON-only constraint, as such adaptation would confound model-level effects with prompt-level optimization. A formal *prompt sensitivity* *analysis*—varying the structured-output shape (variable-length estimates array vs. fixed-slot single primary estimate)—is reported in Section 4.7.

3.7. Pipeline Stages

Each model was run 10 times through two stages:

Stage A: Abstract Screening. Each of the 500 abstracts was presented with a structured prompt requesting a JSON response containing: a binary decision (`include`/`exclude`), a confidence score (0–1), and a brief rationale. Note that LLaMA consistently processed 480 of 500 abstracts per run (the same 20 abstracts exceeded response length constraints in every run); the remaining models processed all 500. Total: 6 models $\times$ 10 runs $\times$ 500 abstracts $=$ 30,000 screening calls.

Stage B: Data Extraction. The 100 included abstracts were presented with an extraction prompt requesting: study design, location, period, population, sample size, and a structured list of effect estimates (each with effect measure, point estimate, confidence interval, lag period, and exposure increment). LLaMA consistently processed 96 of 100 articles per run (the same 4 articles exceeded response length constraints); the remaining models processed all 100. Total: 6 models $\times$ 10 runs $\times$ 100 articles $=$ 6,000 extraction calls.

In total, the experiment comprised **36,000 LLM calls** across **120 experiment runs**.

3.8. Reproducibility Metrics

3.8.1. Exact Match Rate (EMR)

Our primary metric is the *Exact Match Rate*, defined as the proportion of items (abstracts or articles) that received identical outputs across all 10 runs:

$$\mathrm{EMR} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left[\,|\{o_{i,1}, o_{i,2}, \ldots, o_{i,R}\}| = 1\,\right] \tag{1}$$

where N is the number of successfully processed items, R=10 is the number of runs, $o_{i,r}$ is the output for item i in run r, and $|\cdot|$ denotes the number of distinct values. For screening, $o_{i,r}$ is the binary decision; for extraction, $o_{i,r}$ is the SHA-256 hash of the full JSON output (used as a compact identity check: any character-level difference produces a different hash).

An EMR of 1.0 indicates perfect reproducibility; an EMR of 0.0 indicates that every item received at least one different output across runs. We note that extraction EMR is a strict metric—any variation in any field (including formatting differences such as “Beijing, China” vs. “Beijing”) produces a non-match. The field-level analysis in Section 4.4 complements this by isolating which fields drive the variation.

3.8.2. Bootstrap Confidence Intervals

We computed 95% confidence intervals for EMR using the percentile bootstrap method with 10,000 resamples. For each bootstrap sample, we drew N items with replacement and computed the EMR, then reported the 2.5th and 97.5th percentiles of the bootstrap distribution. We note that when EMR=1.000 (as for all local models), the percentile bootstrap trivially returns [1.000, 1.000]. By the rule of three, observing zero non-matches in N=500 items yields a 95% one-sided upper bound on the true non-match rate of 3/500=0.006; thus, we can rule out population-level non-determinism rates above 0.6% for local models, but not lower rates.

3.8.3. Secondary Metrics

- **Flip rate**: 1 – EMR (proportion of items with any variation across runs)

- **Mean agreement**: Average proportion of runs agreeing with the majority decision per item
- **Accuracy vs. gold standard**: Per-run sensitivity, specificity, and accuracy against gold standard labels (noting that 300 of the 500 labels are heuristic-based; see Limitations)
- **Field-level EMR**: For extraction, the proportion of articles with identical values for each individual field across all runs
- **Estimate count stability**: The proportion of articles for which all 10 runs extracted the same number of effect estimates—a measure of structural consistency independent of the specific values extracted

3.9. Provenance Protocol

Scope: client-side observability. The provenance protocol described below is a *client-side* instrument by design. It documents what is observable from the client side—the request payload, the resolved model identifier and version returned by the API (when available), response headers, the client-recorded API response identifier, end-to-end latency, and the input/output text together with their cryptographic hashes. It does not require provider-side weight hashes, GPU-topology fingerprints, or batching-policy attestation, none of which are returned by any production LLM API at the time of writing. The protocol is therefore not restricted to open-source local stacks: for proprietary API-served stacks, it creates the audit trail that providers' opacity would otherwise make impossible. Its operational value is twofold: (i) *differential diagnosis*—when the same prompt produces different outputs at the same client-pinned snapshot, the logged hashes establish that the divergence is attributable to the deployment stack rather than to client-side drift in prompts, inputs, or configuration; and (ii) *provider-agnostic minimum standard*—adopting the protocol creates a verifiable audit trail across any combination of local and API-served stacks that authors may choose to use. Whether providers extend the protocol with weight-checkpoint hashes or deterministic-execution attestation is a separate question of provider commitment; the protocol functions in either case.

Following Rover,[22] we implemented a lightweight provenance protocol that creates a cryptographic fingerprint of every input-output pair, enabling automated detection of any change across runs:

- **Call-level hashing**: SHA-256 hash of a canonicalized JSON object containing five fields (prompt template, input text, model ID, temperature, seed), serialized with sorted keys and ASCII encoding to ensure deterministic hashing across platforms. Absent parameters (e.g., Claude's seed) are encoded as `null`
- **Output hashing**: SHA-256 hash of the raw response text
- **Run cards**: Structured JSON metadata per run, including timestamps, token counts, success/failure rates, and aggregate output hashes

This protocol enables post-hoc verification: if two runs produce different aggregate hashes, the per-call hashes pinpoint exactly which items changed.

3.10. Software and Reproducibility

All code is available at https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility. The experiment was conducted in Python 3.14 on macOS (Apple M4, 24 GB RAM). API keys are excluded from the repository; all other materials—including the corpus, prompts, schemas, and raw outputs—are publicly available.

4. Results

4.1. Overview

All 120 experiment runs completed successfully: 60 screening runs (10 per model × 500 abstracts each) and 60 extraction runs (10 per model × 100 articles each), totaling 36,000 LLM calls. LLaMA consistently processed 480/500 screening abstracts and 96/100 extraction articles across all 10 runs—the same 20 screening and 4 extraction items failed every time due to response length constraints, producing identical error outputs. Because these failures were deterministic (the same items failed identically in every run), they contribute to perfect agreement; EMR was computed over all 500 (screening) and 100 (extraction) items.[1] Mistral, Gemma, Claude, and GPT-4.1 each achieved 500/500 screening and 100/100 extraction in all runs. Gemini achieved 483–500/500 screening and 98–100/100 extraction.

Table 5 summarizes the computational resources consumed. The total experiment required 207 hours of inference time and processed 55.8 million tokens. Local models were substantially slower per call (mean latency 28–92 seconds) than cloud APIs (3–12 seconds), reflecting the difference between single-device Apple M4 inference and distributed GPU clusters. However, the total cost of local inference was negligible ($0.26 electricity) compared to API costs ($69.40), with Claude Sonnet 4.5 accounting for 66% of the total budget.

Table 5. *Computational resources per deployment stack (both stages combined, 10 runs each). Latency is the mean per-call inference duration. Cost includes API pricing and estimated electricity for local stacks (Apple M4, 15W, $0.10/kWh)..*

Deployment Stack	Type	Inference (h)	Tokens (M)	Latency (s)	Cost ($)
LLaMA 3 8B / Ollama / M4	Local	52.1	8.7	32.5	0.08
Mistral 7B / Ollama / M4	Local	76.4	10.6	46.1	0.11
Gemma 2 9B / Ollama / M4	Local	45.6	9.0	27.7	0.07
Claude Sonnet 4.5 / Anthropic	API	7.9	9.9	4.7	46.11
Gemini 2.5 Pro / Google	API	20.3	9.0	12.2	0.00*
GPT-4.1 / OpenAI	API	5.0	8.6	3.0	23.29
Total		**207.2**	**55.8**	—	**69.66**

*Gemini was used under the free tier (rate-limited).

4.2. Screening Reproducibility

Important caveat regarding Mistral 7B (Stage A): Although Mistral 7B achieved EMR=1.000 in the screening stage (Table 6), its specificity against the gold standard was 0.240 and its accuracy was 0.628. Inspection of the outputs reveals that Mistral classified ∼98% of all abstracts as "include"—i.e., a degenerate "always-include" strategy. Any function that returns a constant is trivially reproducible: EMR=1.000 alone is therefore *not* evidence of useful reproducibility. We retain Mistral 7B in our results because this combination (perfect EMR + degenerate specificity) is itself a finding: it warns reviewers that EMR is necessary but not sufficient to characterize a screening tool's utility. We discuss the implications for paired-metric reporting in Section 5.1 of the Discussion.

Table 6 presents the screening reproducibility results.

[1] Computing EMR over only the 480 successfully parsed items yields the same result (EMR=1.000), since the 20 consistently-errored items also agree across runs.

Table 6. *Screening reproducibility across 10 runs per deployment stack. LLaMA's 20 consistently-errored abstracts are included in the EMR denominator (see text). For stacks with EMR=1.000, we report the rule-of-three 95% upper bound on the non-match rate (3/n, Hanley & Lippman-Hand 1983) instead of the trivial bootstrap interval..*

Deployment Stack	**EMR**	**95% UCL***	**Flip Rate**	**Accuracy**	**Sensitivity**	**Specificity**
LLaMA 3 8B / Ollama / M4	1.000	$\leq$0.006	0.0%	0.927	0.928	0.926
Mistral 7B / Ollama / M4	1.000	$\leq$0.006	0.0%	**0.628**	**1.000**	**0.240**‡
Gemma 2 9B / Ollama / M4	1.000	$\leq$0.006	0.0%	0.909	0.969	0.848
Claude Sonnet 4.5 / Anthropic	0.974	[0.960, 0.986]	2.6%	0.931	0.862	1.000
Gemini 2.5 Pro / Google	0.936	[0.914, 0.956]	6.4%	0.964	0.929	1.000
GPT-4.1 / OpenAI	0.970	[0.954, 0.984]	3.0%	0.941	0.882	1.000

*For EMR=1.000: 95% rule-of-three upper bound on non-match rate, $3/n$=3/500=0.006. For EMR<1.000: percentile bootstrap CI.

‡Mistral 7B exhibits a degenerate "always-include" strategy (sens=1.000, spec=0.240). EMR=1.000 is trivial here; see warning box.

All three local models—LLaMA, Mistral, and Gemma—produced identical screening decisions across all 10 runs for every successfully processed abstract (EMR=1.000), confirming that single-device deterministic inference eliminates run-to-run variation entirely. Among the cloud APIs, GPT-4.1 showed a 3.0% flip rate (15 abstracts), Claude 2.6% (13 abstracts), and Gemini 6.4% (32 abstracts).

Notably, accuracy and reproducibility behaved as independent properties: Gemini achieved the highest accuracy (0.964) but the lowest reproducibility (EMR=0.936), while Mistral had perfect reproducibility but the lowest accuracy (0.628) due to an include-biased strategy (sensitivity=1.000, specificity=0.240). This illustrates that a model can be accurate on average yet inconsistent across individual runs, and conversely, that perfect reproducibility does not imply high accuracy.

Inter-run agreement (Fleiss' κ). As a chance-corrected complement to EMR, we computed Fleiss' κ across the 10 runs on the 3-class screening decision (include/exclude/uncertain) for each deployment stack. Inter-run Fleiss' κ for screening ranged from 0.974 (GPT-4.1, "almost perfect") to 1.000 (all three local stacks); the cloud APIs were κ=0.981 (Claude), 0.975 (Gemini), and 0.974 (GPT-4.1). These values quantify self-consistency across repetitions of the same stack and are not directly comparable to the Cochrane $\kappa \geq 0.80$ guideline for agreement between two human reviewers,[1] which measures a different construct (see Section 5.1). For whole-output extraction (each unique output hash as a distinct rating category), the picture is dramatically different: Claude's inter-run κ=0.180 ("slight" on Landis & Koch's scale), Gemini's κ=0.505 ("moderate"), and GPT-4.1's κ=0.512 ("moderate"); local stacks retain κ=1.000. Full Fleiss' κ values with 95% CIs and per-categorical-field decompositions are reported in Supplementary Table 19.

4.3. Extraction Reproducibility

The extraction results reveal a dramatically larger reproducibility gap (Table 7).
While screening EMR remained above 0.93 for all models, extraction EMR dropped dramatically for all three cloud APIs: **0.050 for Claude** (only 5 of 100 articles produced identical extraction outputs across all 10 runs), **0.150 for GPT-4.1** (15 of 100), and **0.200 for Gemini** (20 of 100). All three local models again achieved perfect extraction reproducibility (EMR=1.000). GPT-4.1 showed notably higher estimate count stability (0.922) than Claude (0.880) or Gemini (0.735), indicating that while its outputs varied, the structural composition of extractions was more consistent.

Table 7. *Extraction reproducibility across 10 runs per deployment stack. The "Pairwise" column reports the mean fraction of pairs (across $\binom{10}{2}$=45 run-pairs per article) producing different outputs—a more powerful indicator than EMR alone of how dispersed flipping items are. For EMR=1.000 cells we report the rule-of-three upper bound (3/100=0.030)..*

Deployment Stack	EMR	95% UCL*	Pairwise	Estimate Stability
LLaMA 3 8B / Ollama / M4	1.000	≤0.030	0.0000	1.000
Mistral 7B / Ollama / M4	1.000	≤0.030	0.0000	1.000
Gemma 2 9B / Ollama / M4	1.000	≤0.030	0.0000	1.000
Claude Sonnet 4.5 / Anthropic	0.050	[0.010, 0.090]	**0.818**	0.880
Gemini 2.5 Pro / Google	0.200	[0.120, 0.280]	0.493	0.735
GPT-4.1 / OpenAI	0.150	[0.080, 0.220]	0.486	0.922

*95% rule-of-three upper bound on non-match rate when EMR=1.000.

We emphasize that extraction EMR is computed over the full JSON output hash, so any character-level variation counts as a mismatch. To assess practical significance, we examine field-level variation below.

4.4. Field-Level Analysis

Not all extraction fields are equally affected (Table 8).

Table 8. *Field-level EMR for extraction outputs across deployment stacks. The first three columns are local Ollama/M4 stacks; the last three are API-served (Anthropic, Google, OpenAI)..*

Field	LLaMA	Mistral	Gemma	Claude	Gemini	GPT-4.1
Study design	1.000	1.000	1.000	0.990	1.000	0.990
Study period	1.000	1.000	1.000	0.990	1.000	0.990
Sample size	1.000	1.000	1.000	0.970	1.000	0.980
Population	1.000	1.000	1.000	0.960	0.948	0.990
Study location	1.000	1.000	1.000	0.780	0.896	1.000
Est. stability	**1.000**	**1.000**	**1.000**	**0.880**	**0.735**	**0.922**

A clear pattern emerges: **categorical fields** (study design, study period) are highly reproducible even for API models (EMR≥0.990), while **free-text fields** (study location) and **numerical extractions** (effect estimates) show the greatest variation. GPT-4.1 is notable for achieving perfect location reproducibility (1.000) while showing variation in categorical fields like study design and period (0.990 each)—a different instability profile from Claude and Gemini, where location is the most variable field.

The estimate count stability metric reveals that 26.5% of Gemini's articles with extractable estimates, 12.0% of Claude's, and 7.8% of GPT-4.1's received a different number of effect estimates across runs.[2] This structural variation directly affects any downstream meta-analysis.

[2]Estimate count stability is computed over articles that produced at least one estimate in all 10 runs, yielding denominators of 83 (Gemini), 92 (Claude), and 90 (GPT-4.1) rather than the full 100. The meta-analytic "Varying" column in Table 10 uses the full 100-article denominator, producing slightly lower percentages (20%, 21%, 5%).

4.5. Semantic Equivalence Analysis

The hash-based EMR reported in Table 7 (0.050 for Claude, 0.150 for GPT-4.1, 0.200 for Gemini) treats any character-level difference in the *full JSON output*—including all numeric effect estimates, confidence intervals, and variable-length estimate arrays—as a mismatch. To disentangle variation in the five metadata fields from variation in numeric estimates, and to assess whether metadata variation is merely cosmetic (formatting, abbreviation) or genuinely semantic, we applied three progressively relaxed equivalence criteria to the metadata fields only (Table 9).

1. **Exact match**: character-identical outputs (equivalent to SHA-256 hash comparison).
2. **Normalized match**: after lowercasing, whitespace normalization, punctuation stripping, and abbreviation expansion (e.g., "U.S." → "United States").
3. **Fuzzy match**: Levenshtein similarity ratio ≥ 0.90 between all normalized pairwise comparisons across runs.
4. **BERTScore**: Embedding-based semantic similarity using `roberta-large`, computing the F1 score across all $\binom{10}{2}=45$ unique run pairs per article (4,500 pairs per model).

Table 9. *Semantic equivalence under progressively relaxed criteria ($n = 100$ articles, 10 runs). Columns 2–4 report whole-output EMR over the five metadata fields plus estimate count stability. BERTScore F1 (`roberta-large`) reports the mean ± SD across all $\binom{10}{2}=45$ unique run pairs per article (4,500 pairs per model). Raw F1 is reported here; rescaled F1 (against the package's random-pair English baseline) is shown in Supp. Table 17. Hash-based EMR (from Table 7) additionally requires identical numeric estimates. The gap between high BERTScore (≥ 0.997) and low hash-based EMR (≤ 0.200) demonstrates that even near-identical metadata co-occurs with divergent numeric extractions..*

Deployment Stack	Exact	Norm.	Fuzzy	BERTScore F1†	Hash*
LLaMA 3 8B / Ollama / M4	1.000	1.000	1.000	1.000 ± 0.000	1.000
Mistral 7B / Ollama / M4	1.000	1.000	1.000	1.000 ± 0.000	1.000
Gemma 2 9B / Ollama / M4	1.000	1.000	1.000	1.000 ± 0.000	1.000
Claude Sonnet 4.5 / Anthropic	0.640	0.680	0.690	0.997 ± 0.009	0.050
Gemini 2.5 Pro / Google	0.620	0.620	0.630	0.997 ± 0.022	0.200
GPT-4.1 / OpenAI	0.880	0.880	0.880	1.000 ± 0.004	0.150

*From Table 7; includes all fields and numeric estimates.

†Raw F1 reported; rescaled F1 (against random-pair baseline) shown in Supp. Table 17.

Four findings emerge from this multi-level analysis.

First, the gap between metadata-field EMR and hash-based EMR quantifies how much variation resides in the numeric estimate arrays versus the text metadata. For Claude, 64% of articles had identical metadata across all runs, but only 5% had identical *complete* outputs; for GPT-4.1, 88% had identical metadata but only 15% had identical complete outputs; for Gemini, 62% and 20% respectively. In all three cases, the variable-length estimate arrays (effect sizes, confidence intervals) are the primary source of non-reproducibility.

Second, within the metadata fields, normalization and fuzzy matching improved EMR by only 1–5 percentage points (Claude: 0.640 → 0.690; Gemini: 0.620 → 0.630; GPT-4.1: 0.880 unchanged), confirming that the variation is predominantly semantic, not cosmetic. GPT-4.1's metadata EMR remaining unchanged under normalization and fuzzy matching (0.880) indicates that when its metadata fields vary, the variation is genuinely semantic rather than attributable to formatting differences.

Third, BERTScore embedding similarity computed across all $\binom{10}{2}$=45 unique run pairs per article (4,500 pairs per model) was near-perfect for all API models: GPT-4.1 averaged F_1=1.000 ± 0.004, Claude F_1=0.997 ± 0.009, and Gemini F_1=0.997 ± 0.022, with 100%, 99.7%, and 98.5% of pairs exceeding $F_1 \geq 0.95$, respectively. This creates an apparent paradox: the models produce *semantically near-identical* metadata text, yet 85–95% of their complete extraction outputs are non-identical. The resolution lies in the numeric estimates—the effect sizes, confidence intervals, and variable-length estimate arrays that drive hash-based EMR to 0.050–0.200. Even when a model describes a study in nearly the same words, it may extract a different number of effect estimates or parse "RR=1.05 (95% CI: 1.01–1.09)" with subtly different numeric precision. Concrete, verbatim side-by-side excerpts of an exact-match local-stack output and a near-miss API-stack output (with the divergence located in the variable-length `estimates` array rather than in the metadata fields) are reproduced as JSON listings in Supplementary §5.

Fourth, the three API models exhibit distinct instability profiles. GPT-4.1 shows the highest metadata stability (exact EMR=0.880) and the highest BERTScore (F_1=1.000), yet its hash-based EMR (0.150) confirms that numeric extraction drives the residual variation. Gemini shows the lowest minimum BERTScore (F_1=0.754) and highest standard deviation (0.022), indicating that some articles receive genuinely divergent interpretations across runs—not merely numeric rounding differences. Claude occupies a middle ground in metadata stability but has the lowest hash-based EMR (0.050), suggesting the greatest instability in its numeric extraction layer.

4.6. Meta-Analytic Propagation

To illustrate how extraction non-determinism propagates into downstream analysis, we performed a fixed-effect inverse-variance meta-analysis on the extracted risk ratios (RR/OR/HR) from each run independently (Table 10).

Table 10. *Meta-analytic inputs across 10 runs. "k" is the number of usable effect estimates per run; "k_{sig}" is the number of those estimates whose confidence interval excludes the null. Δk and Δk_{sig} report the range across runs. The "Varying" column reports the proportion of articles for which the number of extracted estimates changed across runs..*

Deployment Stack	**k range**	**Δk**	**$k_{\mathbf{sig}}$ range**	**$\Delta k_{\mathbf{sig}}$**	**Varying**
LLaMA 3 8B / Ollama / M4	246–246	0	188–188	0	0/100 (0%)
Mistral 7B / Ollama / M4	229–229	0	186–186	0	0/100 (0%)
Gemma 2 9B / Ollama / M4	203–203	0	157–157	0	0/100 (0%)
Claude Sonnet 4.5 / Anthropic	186–209	23	152–175	23	21/100 (21%)
Gemini 2.5 Pro / Google	154–171	17	133–150	17	20/100 (20%)
GPT-4.1 / OpenAI	176–180	4	150–154	4	5/100 (5%)

Three findings emerge. **First**, all three local models produced identical estimate sets across all runs: LLaMA extracted 246 estimates, Mistral 229, and Gemma 203 per run, with zero variation—confirming that deterministic inference extends to the structural composition of extractions, not just field-level content.

Second, the three API models showed varying degrees of structural instability. Claude's per-run estimate count varied from 186 to 209 (a 12.4% swing, 21% of articles affected), Gemini's from 154 to 171 (11.0%, 20% of articles), and GPT-4.1's from 176 to 180 (2.3%, 5% of articles).

GPT-4.1 was the most structurally stable API model, with only 5 articles producing a different number of estimates across runs—compared to 20–21 for Claude and Gemini.

Third, the variation in estimate counts directly mirrors the variation in significant estimates. For Claude, the number of significant estimates ranged from 152 to 175 (a swing of 23), meaning that **up to 23 study-level effect estimates—each representing a distinct epidemiological finding—appear or disappear from the evidence base depending solely on which run of the LLM was used**. GPT-4.1's significant estimates ranged from 150 to 154 (a swing of 4), while Gemini's ranged from 133 to 150 (a swing of 17).

While the overall pooled effect in our corpus remained stable across runs for all models (because the aggregate signal is strong and k is large), the instability of the underlying evidence composition is concerning. In domains with smaller literatures or marginal effects, even GPT-4.1's relatively modest variation (Δk=4) could plausibly shift pooled conclusions, while Claude's and Gemini's larger swings pose a more direct threat to meta-analytic reproducibility.

Random-effects re-analysis and small-literature simulation. We re-ran the propagation analysis using DerSimonian–Laird random-effects pooling on log(RR), one pooled estimate per model×run combination over the 100 INCLUDE articles. Across all 60 combinations, the pooled RR ranged from 1.015 to 1.038 (mean 1.025); the within-model range across runs was small in absolute terms (Claude 0.0058, Gemini 0.0058, GPT-4.1 0.0058) and remained within the bootstrap CI of any single run. **No run-pair within any model produced a 95% random-effects CI that crossed the null** in the full-corpus pooling, confirming that, in a well-powered, large-k meta-analysis with a strong aggregate signal, LLM non-determinism does not flip the meta-analytic conclusion.

This robustness, however, depends critically on k. We subsampled the corpus to plausible small-literature scenarios ($k \in \{10, 15, 20, 30\}$, N=200 random subsamples per k, seed=42) and computed random-effects pooled estimates for each of the 10 LLM runs. For k=10 subsamples, **0.5% of Claude and Gemini subsamples** (1 of 200 each) produced *unstable null-crossing* under the DerSimonian-Laird Wald-z formulation: the pooled 95% CI crossed RR=1 in some runs but not others, meaning that the meta-analytic conclusion (significantly above null vs. not significant) **changed depending on which LLM run generated the inputs**. For $k \geq 15$, no subsample exhibited unstable null-crossing across runs in any model under DL. This demonstrates a concrete, plausible scenario where LLM non-determinism affects the downstream conclusion: emerging literatures with $k \sim 10$ articles (small reviews of new exposures, rare outcomes, or recent topics) are susceptible to run-dependent reversals, even when the underlying signal is detectable.

Because DerSimonian-Laird is known to be liberal in small-k settings, we re-ran the simulation with the Hartung-Knapp-Sidik-Jonkman (HKSJ) variance correction (Cochrane Handbook 6.5+ recommendation for $k<10$). Under HKSJ at k=10, the unstable-null-crossing rate rises to **31.5% (Claude), 48.5% (Gemini), and 20.0% (GPT-4.1)**, with non-trivial rates persisting up to k=20 for Claude and Gemini (16.5% and 10.5%, respectively) before vanishing at k=30. The qualitative conclusion is robust to the variance estimator (run-dependent reversals affect only small literatures); the magnitude is substantially amplified under the more conservative HKSJ formulation appropriate for $k<10$. Full DL-vs.-HKSJ comparison appears in Supplementary Section 19 and Table 21.

4.7. Fixed-Slot Extraction Sensitivity

To test whether the variable-length `estimates` array structurally amplifies cloud API non-determinism, we re-ran the extraction stage on the three cloud models with a fixed-slot prompt that requests *exactly one* primary estimate (lag 0–1 if multiple lags are reported, single-pollutant model preferred, all-respiratory preferred over subgroups, normalized to per 10 μg/m^3). All other parameters were held identical to the variable-length design: temperature=0, seed=42, same 100 INCLUDE articles, three runs per model.

Table 11. *Fixed-slot extraction EMR vs. variable-length baseline (3 runs vs. 10 runs; 100 INCLUDE articles)..*

Deployment Stack	Var-length EMR	Fixed-slot EMR	Relative change
Claude Sonnet 4.5 / Anthropic	0.050	0.030	−40.0%
Gemini 2.5 Pro / Google	0.200	0.185	−7.6%
GPT-4.1 / OpenAI	0.150	**0.444**	**+196.3**%

The result is *not* a uniform improvement. GPT-4.1's EMR almost tripled with the fixed-slot prompt (0.150→0.444), confirming that for that model, prompt structure—specifically the variable-length array—was a major contributor to non-determinism. Claude actively *worsened* (−40%); Gemini was effectively neutral. This refines the expectation, voiced in peer review, that "fixed-slot extraction will resolve cloud non-determinism": the effect of prompt design is real but **model-specific**, and for at least Claude the dominant source of variation lies elsewhere (likely in the cloud serving stack itself, see Section 4.8).

4.8. Deployment-vs.-Model-Size Desconfound Experiment

Throughout the comparisons above, the local versus cloud difference is confounded with model size: our three local models are 7–9B-parameter open-weight models, while the three cloud APIs are estimated at >100B parameters and trained with proprietary procedures. To isolate the deployment effect, we ran a seventh experimental condition: the identical `meta-llama/llama-3-8b-instruct` model—same weights, same tokenizer, same architecture—served via cloud (OpenRouter routing pinned to the DeepInfra provider) at temperature=0 and seed=42. Ten cloud runs × 500 abstracts mirror the local experimental design.

The two arms hold constant the model weights, the prompt source file, the decoding temperature and the seed, but they do not issue a byte-identical request payload, and we state the difference rather than let the word "identical" carry it. The local harness interpolates the article into the prompt template before the call and the serving layer then appends the article text again, whereas the cloud harness transmits the uninterpolated template and the article as separate message parts; the two were additionally configured with different output-token ceilings (2,048 local, 512 cloud). Measured over the deposited call records, the arms differ by 34.2% in mean input tokens (1,279.3 local vs. 842.3 cloud) and share *none* of the 500 canonical call hashes. This condition therefore holds the weights constant while varying the remainder of the deployment stack *including its request-rendering layer*, and we scope the inference accordingly: it demonstrates that identical weights do not guarantee identical behaviour across serving stacks, and it does not isolate the serving infrastructure from the payload construction that stack imposes. Output truncation is excluded as a contributor by direct measurement: all 4,266 completed cloud calls returned `finish_reason=stop`, none hitting the 512-token ceiling.
Three findings stand out.

First, infrastructure introduces non-determinism in a small-parameter model. Local LLaMA 3 8B is bit-deterministic (EMR=1.000 over 10 runs and 500 abstracts). The same model served via cloud is *not*: 2.04% of items receive different decisions in at least one run-pair (mean pairwise disagreement 0.79%). This refutes the model-size confound directly: a

Table 12. *Same-weights, different-stack comparison. Two deployment stacks share the `meta-llama/llama-3-8b-instruct` weights but differ in the remaining tuple components: local stack = (weights, Meta/open, Ollama v0.15.5/Apple M4/single replica, local HTTP); cloud stack = (weights, Meta-via-DeepInfra, OpenRouter routing/opaque GPU pool, OpenRouter Chat Completions). Ten runs each, same prompt source file, seed=42, temperature=0; the request payloads are not byte-identical (see text)..*

Quantity	**Local stack (Ollama / M4)**	**Cloud stack (DeepInfra)**
EMR within deployment	1.000	0.992*
Mean pairwise disagreement	0.0000	0.0079
% items with any pair-disagreement	0.00%	2.04%
Cross-deployment agreement (all run pairs)	—	59.7%
Items where local=include / cloud=exclude	—	167
Items where local=exclude / cloud=include	—	**0**

*Pairwise-mean-based estimate; 2 of 10 cloud runs had transient infrastructure failures that reduced their valid-item counts (45 and 221, vs. ~491 in the other 8 runs).

7–9B model can become non-deterministic purely by virtue of its serving stack, with no change in weights, tokenizer, prompt, seed, or temperature.

Second, cloud and local produce systematically different outputs. Even within the runs where the cloud was internally consistent, mean pairwise agreement with the local deployment was 59.7%. Of 430 items where both deployments produced a valid decision, 167 (38.8%) showed a different decision under cloud than under local.

Third, the discordance is asymmetric. Of those 167 discordant items, **all 167** were cases where the local model said *include* but the cloud said *exclude*; the reverse direction (local exclude / cloud include) occurred zero times. The cloud deployment is systematically more conservative. The mechanistic interpretation is that the cloud serving stack uses batched continuous-batching inference with floating-point reduction order that depends on batch composition at the moment of inference; local Ollama uses single-stream sequential inference on a single Apple M4 chip with deterministic reduction order. The asymmetry of the disagreement is suggestive of a systematic bias in how cloud floating-point summation truncates positive evidence, but isolating that mechanism is beyond the scope of this study.

We additionally observed an operational reliability finding: 2 of 10 cloud runs (`run_005`, `run_006`) experienced transient infrastructure failures that reduced their successful-call counts substantially (45/500 and 221/500 versus the ~491/500 baseline of the other eight runs). This pattern was absent from all 10 local Ollama runs, all of which had identical 480/500 success. Cloud reliability for systematic-review work is itself a finding worth reporting alongside reproducibility.

4.9. Summary Comparison

Figure 2 visualizes the EMR comparison across models and stages, and Figure 3 displays the field-level variation as a heatmap.

The results reveal three key patterns: (1) a sharp reproducibility gap between local and API-served deployment stacks, with all three local stacks achieving EMR=1.000 while all three API-served stacks fall below 0.200 for extraction; (2) a dramatic amplification of non-determinism when moving from simple binary screening decisions to complex structured extraction outputs; and (3) distinct instability profiles among API-served stacks, with the GPT-4.1 / OpenAI stack showing the highest metadata and structural stability but still exhibiting meaningful numeric variation.

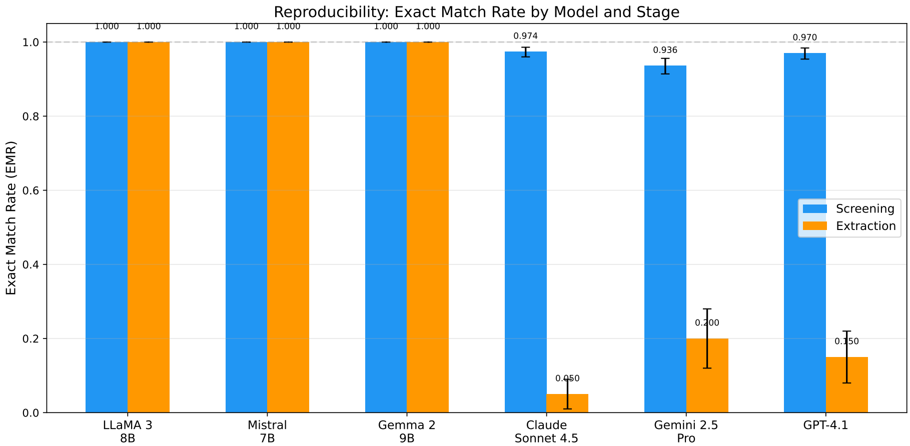

Figure 2. *Exact Match Rate (EMR) by deployment stack and pipeline stage. Six deployment stacks are shown: three local Ollama/Apple M4 stacks (LLaMA 3 8B, Mistral 7B, Gemma 2 9B) and three API-served stacks (Claude Sonnet 4.5/Anthropic, Gemini 2.5 Pro/Google, GPT-4.1/OpenAI). Error bars represent 95% percentile bootstrap confidence intervals over 10,000 resamples; for stacks with EMR=*1.000*, the displayed upper bound is the rule-of-three* $3/n$ *instead of the trivial bootstrap interval. The dashed line indicates perfect reproducibility (EMR=*1.0*)..*

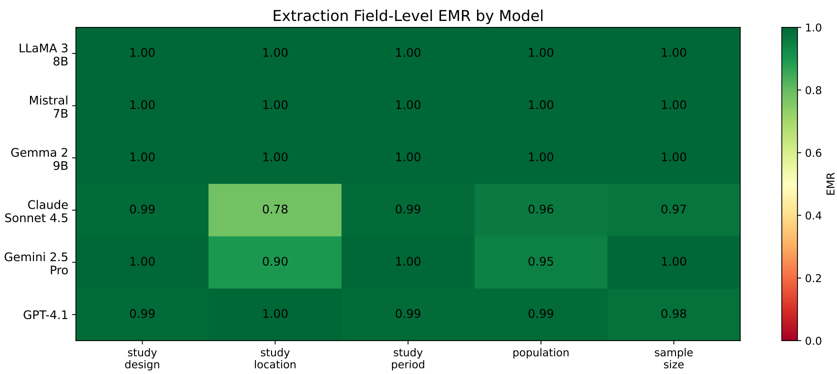

Figure 3. *Field-level EMR heatmap for extraction outputs across the six deployment stacks (three local Ollama/Apple M4 stacks on the left columns; three API-served stacks—Claude/Anthropic, Gemini/Google, GPT-4.1/OpenAI—on the right). Darker cells indicate lower reproducibility. Categorical fields (study design, study period) remain stable across all stacks, while free-text fields (study location) and the variable-length structured estimates array show the greatest variation in the three API-served stacks..*

5. Discussion

5.1. Principal Findings

Our study provides a systematic quantification of LLM non-determinism across a complete evidence synthesis pipeline—to our knowledge, the first to combine whole-output metrics, multi-run repetition, and pipeline propagation analysis in this domain. Four key findings emerge.

Reframing: the object of reproducibility variation is the deployment stack, not the model. A first-principles consequence of our results is that "which model" is the wrong level of abstraction for talking about LLM reproducibility. A deployment stack—the tuple of weights, provider, infrastructure, and API layer (Section 3.2)—is the unit through which an LLM actually generates an answer, and it is the unit at which reproducibility is preserved or destroyed. Two stacks that share weights can diverge by tens of percentage points if their infrastructure or serving layer differs (Table 12); two stacks that differ in weights but share a deterministic local infrastructure can both achieve EMR=1.000 (Tables 6, 7). Reporting only the model name in an LLM-assisted SR therefore underspecifies the experimental condition: the same weights on a different stack will not reliably reproduce the same outputs. This reframing aligns our results with a broader programme on reproducibility in API-served LLMs developed in a companion paper (Rover & Tadano, under review at *Nature Communications*) and motivates the recommendations in Section 5.6.

Finding 1: Deployment infrastructure—not parameter count—drives the local-vs-cloud reproducibility gap. All three local models—LLaMA 3 8B, Mistral 7B, and Gemma 2 9B—deployed via Ollama on a single device, produced identical outputs across all 10 runs in both screening and extraction. All three cloud-based APIs (Claude, Gemini, GPT-4.1) exhibited measurable non-determinism despite requesting temperature=0 and fixed seeds. The desconfound experiment in Section 4.8 establishes that this gap is *not* attributable to model size: serving the identical `meta-llama/llama-3-8b-instruct` via cloud (DeepInfra) introduces 2% pairwise disagreement and 39% systematic disagreement with the local deployment of the same weights—asymmetrically, with the cloud always more conservative. This gap has a precise technical explanation rooted in floating-point arithmetic. Cloud APIs serve requests across multi-GPU clusters where matrix multiplications are partitioned and reduced in non-deterministic order. Because floating-point addition is not associative—$(a+b)+c \neq a+(b+c)$ in general due to rounding—different partitioning strategies produce subtly different intermediate sums, which propagate through the autoregressive generation process and can eventually cause different token selections.[21] Local deployment on a single device with a single execution thread avoids this source of variation entirely: the same computation graph executes in the same order every time, producing bit-identical results. This determinism guarantee is configuration-specific, however. The same model on different hardware, driver versions, or Ollama versions would likely produce different baseline outputs. Determinism holds within each configuration, not across configurations.

Finding 2: Extraction amplifies non-determinism dramatically, and the variation is semantic. Claude's EMR dropped from 0.974 (screening) to 0.050 (extraction); GPT-4.1's from 0.970 to 0.150; Gemini's from 0.936 to 0.200. Screening requires a binary decision (a single token effectively determines the output), while extraction generates complex structured JSON with multiple fields. Our four-tier semantic equivalence analysis (Section 4.5) confirms that this gap is not an artifact of strict string matching. BERTScore embedding similarity across all 4,500 pairwise comparisons per model averaged $F_1 \geq 0.997$—near-perfect semantic agreement on metadata text—yet hash-based EMR remained at 0.050–0.200. This paradox reveals that the variation concentrates in the numeric extraction layer: the models describe studies in nearly the same words but extract different numbers of effect estimates and parse numeric values with different precision. Even the residual metadata variation is genuinely semantic (fuzzy matching recovers only 0–5 percentage points), not cosmetic. The practical implication depends on context: the high BERTScore ($F_1 \geq 0.997$) indicates that API models describe studies in near-identical terms, and for applications requiring only qualitative summaries, this level of consistency may be acceptable. However, for quantitative evidence synthesis—where exact numeric values feed directly into meta-analytic pooling—the hash-based EMR of 0.05–0.20 represents a genuine threat to reproducibility.

Finding 3: API-served stacks exhibit distinct instability profiles. GPT-4.1 achieved the highest metadata stability among API models (exact EMR=0.880, BERTScore F_1=1.000) and the smallest structural variation (Δk=4 estimates across runs), yet its hash-based extraction EMR (0.150) confirms that numeric variation persists. Claude had the lowest hash-based EMR (0.050) and the widest structural swing (Δk=23), while Gemini showed the most extreme semantic divergence (minimum BERTScore F_1=0.754). These distinct profiles suggest that non-determinism manifests differently across API providers—some vary primarily in numeric precision, others in structural composition, and others in semantic interpretation.

Finding 4: Accuracy and reproducibility are independent properties; report them as a paired tuple. Gemini achieved the highest screening accuracy (0.964) but the lowest screening reproducibility (0.936), while Mistral had perfect reproducibility (EMR=1.000) but a degenerate "always-include" strategy (specificity=0.240, accuracy=0.628). The Mistral case is a stark warning: a constant function is trivially reproducible, so EMR alone is not evidence of useful reproducibility. This independence means that **accuracy evaluations from a single run cannot detect reproducibility problems**, and **reproducibility evaluations without accuracy cannot detect degenerate strategies**. Future studies should report EMR and accuracy (against a human-labeled ground truth) as a paired tuple, never EMR alone.

Finding 5: Prompt design is a contributor to extraction non-determinism, but the effect is model-specific. The fixed-slot extraction sensitivity analysis (Section 4.7) showed dramatic improvement for GPT-4.1 (EMR 0.150→0.444, +196%), but no improvement—or active worsening—for Claude (−40%) and Gemini (−7.6%). This refines the assumption that "fixed-slot will fix it": prompt structure matters, but the dominant source of variation differs across providers. For Claude in particular, the residual non-determinism after prompt simplification points back to deployment-stack variation rather than prompt-induced array variability.

Finding 6: The seed parameter delivers a smaller effect than the deployment paradigm. The journal's own guidance for GenAI submissions requires authors to document specific random-seed values and, for commercial models, API versions and access dates;[17] it stops short of establishing whether a seed parameter delivers determinism once documented. Our data permit a direct test: Gemini and GPT-4.1 accept a seed parameter (set to 42); Claude does not. For screening, seeded cloud models had EMR slightly *below* non-seeded Claude (0.953 vs. 0.974); for extraction, seeded models were better than Claude by 0.125 EMR. The deployment effect—local seeded EMR=1.000 vs. cloud seeded mean EMR=0.175 in extraction—is **6× larger** than the seed effect (cloud-seeded vs. cloud-unseeded EMR difference of 0.125). We conclude that the seed parameter, as currently implemented in commercial APIs, does not deliver the determinism that users may infer from its presence; deployment paradigm dominates as the primary controllable variable.

5.2. Implications for Evidence Synthesis Practice

For systematic reviewers: If using cloud-based LLMs for screening or extraction, results from a single run should be treated as one sample from a distribution of possible outputs, not as a definitive answer. Running multiple repetitions and reporting variation metrics should become standard practice, analogous to how human inter-rater reliability is routinely measured. It is tempting to place the models' screening EMR of 0.93–0.97 next to the Cochrane expectation of $\kappa \geq 0.80$ for human screening agreement[1] and read the comparison as favourable, and we resist that reading for two reasons. First, the quantities are not commensurable: EMR and inter-run Fleiss' κ measure how consistently a *single* rater reproduces itself across repetitions, whereas Cohen's κ measures whether *two different* raters reach the same conclusion. A system can be perfectly self-consistent and consistently wrong—in this corpus Mistral-7B achieves

EMR=1.000 at a specificity of 0.240 (Section 4.4), which is the cleanest possible demonstration that self-consistency is not accuracy. Second, on this same corpus our own two human raters reached κ=0.529 (95% CI [0.383, 0.674]; Section 3.3), so the 0.80 figure functions here as an aspirational guideline rather than as a measured property of human screening. The honest comparison is therefore not "models versus the threshold" but "models and humans each measured, on different constructs, against a threshold that neither was shown to meet." Extraction EMR (0.05–0.20), by contrast, requires no such careful framing: it falls below any reliability standard on any construct.

For journal editors and reviewers: Papers reporting LLM-assisted evidence synthesis should be expected to disclose: (a) the exact model version, (b) all inference parameters, (c) whether results were verified across multiple runs, and (d) provenance information enabling independent replication.

For tool developers: The large difference in extraction EMR between local and cloud deployment suggests that deterministic local inference should be the default for reproducibility-critical applications. Local inference on consumer hardware (Apple M4, 24 GB RAM) cost $0.26 total for 174 hours across three models, versus $69.40 for 33 hours of API inference—a 267× cost advantage per hour that compounds with the reproducibility benefit. Where cloud APIs must be used (e.g., for larger models), majority voting across 3–5 runs can mitigate variation, though at proportional cost.

5.3. Relationship to Prior Work

Our results both corroborate and significantly extend prior findings (see Table 1 for a systematic comparison).

Our screening flip rates (2.6% for Claude, 3.0% for GPT-4.1, 6.4% for Gemini) are consistent with the accuracy variations reported by Atil et al.[18] on general benchmarks, confirming that the phenomenon documented in controlled settings translates directly to real-world evidence synthesis tasks. However, we reveal that **structured extraction amplifies non-determinism by an order of magnitude**—a finding invisible to benchmark-level analyses. Defining amplification as the ratio of extraction flip rate to screening flip rate, Claude's non-match rate increased from 2.6% to 95.0% (36.5×), GPT-4.1's from 3.0% to 85.0% (28.3×), and Gemini's from 6.4% to 80.0% (12.5×). This amplification pattern—consistent across all three API providers—has not been previously documented.

The comparison with Jensen et al.[10] is particularly instructive. Their reported 94.1% inter-session reproducibility for ChatGPT-4o extraction appears to align with our Claude screening EMR (97.4%), creating the impression that API-based extraction is reasonably stable. However, their metric measures per-field agreement across 2 sessions, while our EMR measures whole-output identity across 10 runs. When we apply our stricter metric, extraction reproducibility drops to 5.0%—**nearly twenty times lower than what per-field metrics suggest**. This discrepancy is not an artifact; it reflects a genuine measurement gap. Per-field metrics miss the combinatorial explosion of variation: when 5 fields each have 97% individual reproducibility, the probability that *all* fields match simultaneously is only $0.97^5 = 0.859$—and this is before accounting for the variable-length effect estimate lists that drive most of our observed non-matches.

Similarly, Gartlehner et al.'s[9] 95–97% test-retest reliability with Claude 2 was measured over 2–3 repetitions. Our 10-run design with output hashing reveals instability that shorter test-retest windows cannot detect, because some items change in only 1–2 of 10 runs—a frequency that 2-run comparisons would miss entirely.

The meta-analytic propagation experiment (Section 4.6) provides the most direct evidence of downstream impact. For Claude, the number of usable effect estimates varied from 186 to 209 across runs—a 12.4% swing in the evidence base feeding the pooled result. For

Gemini, the swing was 11.0% (154–171), and even the most stable API model, GPT-4.1, showed a 2.3% swing (176–180). Across API models, 5–21% of articles produced a different number of effect estimates depending on the run, meaning that **the composition of a meta-analysis is not fixed when an LLM extracts the data**. Importantly, the overall pooled conclusion remained stable across all runs in our corpus—a finding we attribute to the large k and strong aggregate effect, which buffer against compositional variation. This stability should not be misread as evidence that non-determinism is inconsequential; rather, it reflects the deliberately robust design of our evaluation corpus. In systematic reviews targeting smaller literatures ($k<20$) or marginal effects, even GPT-4.1's modest variation ($\Delta k=4$) could shift a confidence interval across the null. We used a fixed-effect inverse-variance model; a random-effects model (standard for heterogeneous epidemiological studies) would produce wider confidence intervals, making the pooled conclusion *more* sensitive to compositional changes—further underscoring the importance of multi-run verification.

5.4. Cost-Benefit Considerations for SR Teams

A practitioner considering our findings faces a deployment trade-off between (a) local open-weight serving with full reproducibility but slower throughput, and (b) cloud-API serving with high throughput but stochastic outputs. We sketch a quantitative comparison for a typical SR team conducting 5 reviews per year, each with a 2,000-abstract screening pool and a 100-article extraction stage.

Scenario A — Local-pinned (Gemma 2 9B with pinned Ollama version + pinned hardware). Single-pass screening of 2,000 abstracts takes ≈14 hours per review on Apple M4 (25.2 sec/abstract, our measured rate). Annual screening time: 70 hours. Extraction: $100 \times 5 \times 70$ sec = 9.7 hours. Total annual LLM time: 80 hours. Equipment cost: amortized hardware + electricity (estimated US$300/year at 24/7, US$0.12/kWh). Reproducibility: bit-deterministic.

Scenario B — Cloud + 3×-run majority vote (GPT-4.1). Single-pass screening: ≈1.4 hours; 3× runs = 4.2 hours per review. Annual screening: 21 hours. Extraction with 3× vote: ≈1.7 hours. Total LLM time: 22.7 hours. API cost: ≈US$15 screening + US$8 extraction per review = US$115/year. Reproducibility: high but not bit-deterministic (3× majority captures ≥99% of stable outputs at the cost of 3× API spend).

Scenario C — Cloud single-run (typical current practice). Same as B but with one run. Total LLM time: 7.6 hours; cost ≈US$38/year. Reproducibility: not guaranteed; replication attempts will produce different outputs in 5–20% of articles.

For teams that value rigorous reproducibility and have a single moderate workstation, Scenario A is cost-effective and guarantees identical outputs across re-runs. For throughput-prioritizing teams willing to accept documented non-determinism mitigated by majority voting, Scenario B saves roughly 60 hours of screening labor per year at ∼US$115. A hybrid (local screening + cloud extraction with voting) is also feasible.

5.5. Structured-Output and Tool-Use Modes as Potential Mitigations

The three commercial APIs in our study all offer "structured output" or "tool-use" modes that constrain output format more strictly than free-form JSON: Anthropic's tool-use API enforces a Pydantic-like schema server-side, OpenAI's `response_format=json_schema` similarly constrains the output token distribution, and Gemini's function-calling enforces argument types in the response.

Whether these modes reduce run-to-run non-determinism in extraction is an open question that our study did not directly address—we used the standard Chat Completions endpoints with prompt-level JSON instruction, matching the most common practice in the LLM-SR literature as of late 2025. Two hypotheses are testable in future work. First, structured outputs may reduce *parsing* variation (no more "Markdown code-fence around JSON" edge cases) without reducing the underlying semantic non-determinism (the model still chooses among multiple valid outputs). Second, structured outputs with strict enums (e.g., `lag` restricted to `lag0, lag1, lag0-1, lag2`) may push semantic variation toward the specific sub-fields with the most uncertainty, but bounded EMR may improve. We recommend that future LLM-SR studies report results in both unstructured and structured-output modes when comparing across commercial APIs, because the deployment paradigm explicitly affects which output channels are available.

5.6. Scope, Design Decisions, and Research Opportunities

Every experimental design involves deliberate scope choices. We document ours transparently—not as weaknesses, but as controlled boundaries that define the contribution and open clearly charted directions for future work.

Controlled stack selection. We deliberately selected six deployment stacks spanning two infrastructure paradigms (local single-device vs. API-served opaque cluster), four providers, and two weight-size regimes (7–9B vs. >100B estimated parameters). This design is the broadest reproducibility comparison in evidence synthesis to date. We note that the local-vs-API axis varies multiple tuple components simultaneously: weights, infrastructure, and serving layer all differ between, e.g., the LLaMA 3 8B / Ollama / M4 stack and the Claude / Anthropic / Messages API stack. The desconfound experiment in Section 4.8 addresses this by holding weights constant while varying the rest of the tuple, but the same logic applied to additional weight-checkpoint pairs (e.g., Mixtral 8x7B served locally and via two different cloud providers) would map the stack-level contribution more comprehensively. The practical finding holds regardless of mechanism: stacks built on local single-device inference *are* deterministic; stacks built on opaque API-served infrastructure *are not*. Extending coverage to additional architectures (Command-R, DeepSeek, Phi-4) and larger open-weight models (70B+) on multiple infrastructure tiers remains a natural next step.

Deliberate run count. Ten repetitions per model-stage combination provide adequate statistical power for EMR estimation with bootstrap confidence intervals, yielding narrow CIs for screening (e.g., ± 0.016 for Claude) and informative CIs for extraction. Our prior work[22] demonstrated that scaling to 100 repetitions enables detection of tail-event instability; future studies targeting rare variation patterns (<10% of runs) can build on our infrastructure by simply increasing the run count.

Domain focus as a strength. We chose $PM_{2.5}$/respiratory health because it provides a well-characterized testbed with standardized reporting conventions (relative risks, confidence intervals, exposure lags), enabling clean measurement of non-determinism without confounding from heterogeneous reporting styles. This domain focus is a feature, not a limitation: it provides a controlled baseline against which cross-domain studies—in behavioral health, pharmacology, or nutritional epidemiology, where reporting is more heterogeneous—can calibrate expected amplification patterns.

Conservative screening EMR. Our screening EMR intentionally measures only the binary include/exclude decision, which is the operationally consequential output. This is a conservative choice: the full output also contains confidence scores, rationales, and sub-classifications that likely vary more than the binary decision. Our reported screening EMR values therefore *underestimate* the true extent of screening-stage non-determinism, making our findings a lower bound.

Stress-tested corpus design. Our corpus was deliberately enriched with 60% ambiguous abstracts to stress-test screening consistency under challenging conditions. In a typical systematic review, borderline cases constitute approximately 20–30% of abstracts, meaning our screening flip rates represent upper bounds for standard workloads. This design choice maximizes the diagnostic sensitivity of our evaluation.

Temporal controls. All experiments were completed within a 3-week window (February 11–March 2, 2026) to minimize the risk of silent API model updates. We pinned Claude to a dated version (`claude-sonnet-4-5-20250929`) and used versioned GPT-4.1. Longitudinal studies tracking EMR over weeks or months represent a compelling research opportunity to distinguish inference-level non-determinism from model-update effects.

Deployment-stack opacity and snapshot validity. For the three API-served stacks, the infrastructure and intermediate-precision routines—GPU topology, batching policy, weight checkpoint version, internal post-training—are opaque to the client; no metadata returned by the API exposes them, and providers may change them silently. Our results therefore characterize the snapshot of each API stack observed during data collection (February–April 2026), and exact response-level reproducibility may not survive future provider-side updates even with identical client requests. The three local stacks, by contrast, expose every tuple component (open weights with verifiable hashes, single Apple M4 chip, Ollama v0.15.5, single replica), which is why the local-vs-API comparison is informative as a partial quasi-isolation probe (Section 3.2): the local side is fully specified and pinned, so any cross-stack divergence localises to non-weight components.

External validation as documented limitation. This study tests reproducibility within a single 500-abstract corpus on $PM_{2.5}$ and respiratory hospitalization. We have not validated our findings on an independent test corpus drawn from a different domain or time period. Three threats follow. First, generalization to other domains: $PM_{2.5}$/respiratory is a well-structured, mature epidemiology literature with formulaic outcome reporting (RR/OR per 10 μg/m^3 with 95% CI, time-series design language); LLM extraction stability may differ for less-structured domains such as qualitative health-services research, behavioral interventions, or nutritional epidemiology. Cross-domain reproducibility is the focus of our companion study (Rover *et al.*, in preparation), which uses an air-pollution/health corpus paired with an artificial-intelligence/ML corpus to isolate domain effects. Second, generalization to other time periods: our experiments executed within a 3-week window (February 11–March 2, 2026) to control for silent model updates; cloud APIs may behave differently in earlier or later periods, so reproducibility-measurement studies have a built-in shelf life. Third, hardware generalization: the local-model EMR=1.000 result was obtained on a single device (Apple M4, 24 GB RAM, macOS Darwin 24.6.0, Ollama v0.15.5). Replication on a second, non-Apple-M4 device (e.g., Linux x86_64 with Ollama under different driver versions) is a natural follow-up; we did not perform this pilot in the present study and document it as future work.

Open research frontiers. Our findings open several high-impact research directions: (1) establishing clinically meaningful equivalence thresholds (what BERTScore difference corresponds to a materially different meta-analytic conclusion?), (2) directly testing significance flipping in smaller literatures (k<20) with marginal effects, where even the modest compositional instability we documented could shift a confidence interval across the null, (3) investigating whether structured-output modes (Anthropic tool-use, OpenAI `response_format=json_schema`, Gemini function-calling) reduce non-determinism without sacrificing accuracy, and (4) mechanistically isolating which component of cloud serving (batched continuous-batching, dynamic GPU/node assignment, floating-point reduction order) drives the asymmetric local-vs-cloud disagreement we documented in Section 4.8.

Pre-registration disclosure. This study is partly pre-registered, and we state precisely which parts, because the distinction matters for how much weight each result can carry. The original analyses (Sections 4.4–4.6) are exploratory: the choice of 10 repetitions, whole-output hash EMR, four semantic-tier definitions, and fixed-effect meta-analytic propagation were made during analysis design but without an Open Science Framework or AsPredicted submission. The dual-human labeling validation is the exception and is *formally* pre-registered: the protocol, the screening templates, and the agreement script were frozen as an OSF registration on 2026-05-12 (https://doi.org/10.17605/OSF.IO/FGN3E), two months before either label set was returned, and that registration pre-specified the $\kappa \geq 0.80$ target together with the contingency followed in Section 3.3 when the target was missed. The remaining post-revision analyses (Sections 4.7–4.8, silver-external standard) were designed and committed to the public Git repository *before* result computation, with commit hashes serving as informal cryptographic timestamps; we do not present Git commits as equivalent to pre-registration. We acknowledge the absence of formal pre-registration for the exploratory core as a limitation and recommend that future LLM-in-SR work register analyses on OSF or AsPredicted before data collection. We did not retrofit pre-registration claims to the original exploratory analyses.

5.7. Recommendations

Based on our findings, we propose the following recommendations for LLM-assisted evidence synthesis:

1. **Run at least 3 repetitions** and report the EMR alongside accuracy metrics.
2. **Prefer local deployment** for reproducibility-critical tasks when model capability permits.
3. **Implement provenance hashing** to enable post-hoc verification and auditing.
4. **Report full model specifications**: model ID, version, temperature, seed, and any API-specific parameters.
5. **Use majority voting** when local deployment is not feasible, accepting the cost trade-off for improved stability.
6. **Distinguish accuracy from reproducibility** in evaluations—a model can be accurate on average but unreliable for any single run.

6. Conclusion

This study evaluates LLM reproducibility in evidence synthesis at scale: 35,996 successful calls of 36,000 attempted across 120 original experimental runs involving six models from four providers, plus a desconfound experiment serving the same model weights via a seventh cloud configuration. We use whole-output hashing, four-tier semantic analysis, pairwise-disagreement statistics, fixed-slot prompt sensitivity, two independent silver standards, and meta-analytic propagation testing including a small-literature simulation.

What we show that others have not. Where prior studies report 94–97% reproducibility using per-field metrics and 2–3 repetitions, our whole-output analysis across 10 runs reveals that only 5–20% of extraction outputs are truly identical—exposing a measurement gap of nearly 20×. Where prior studies measure reproducibility on general benchmarks, we demonstrate that structured evidence extraction *amplifies* non-determinism by 13–37× compared to binary screening—a phenomenon invisible to benchmark evaluations. Where prior studies report either accuracy or reproducibility, we show these properties are *independent*: a model can rank first in accuracy and last in reproducibility.

What we prove is achievable. Deterministic LLM inference is not aspirational—it is already achievable. All three open-weight models (LLaMA 3 8B, Mistral 7B, Gemma 2 9B) produced bit-identical outputs across all 10 runs in both stages on fixed local hardware, at a total cost of US$0.26. The desconfound experiment in Section 4.8 eliminates the model-size hypothesis: serving the identical `meta-llama/llama-3-8b-instruct` weights via cloud (DeepInfra) introduces 2% pairwise disagreement and 39% asymmetric disagreement with the local deployment. The practical implication is therefore stronger than "locally-served open-weight models are deterministic": the determinism comes from the serving stack, not the model. Single-stream sequential inference on a single device with deterministic floating-point reduction order is what produces bit-identical outputs; batched cloud inference with dynamic node allocation does not.

Relationship to companion work. The provenance protocol used in this study is identical to that proposed and validated in our companion paper (Rover & Tadano, 2026, in press at JAIR[22]), which establishes a general-purpose LLM call-hashing standard tested across 8 models on a 100-abstract AI/ML corpus with 100 repetitions per condition. The present paper applies that protocol to a different research question—pipeline-level propagation through a complete two-stage SR workflow—using a smaller per-condition repetition count (10 runs) to enable broader pipeline coverage (screening + extraction + meta-analytic propagation) on a domain-specific corpus. The choice of 10 repetitions here was made to balance pipeline depth (two stages, six models, full meta-analytic re-pooling, plus the desconfound and fixed-slot conditions) against per-condition precision; with 10 runs we have $\geq$95% power to detect a per-item non-match probability of 0.30 or higher (under a binomial model), while rare non-determinism (per-item p<0.05) is under-detected. The companion paper's 100-run design at fewer model-task combinations provides complementary high-precision evidence that this paper's pipeline-coverage design cannot. Readers seeking precise per-item non-match rates should consult the companion paper; readers interested in propagation through a complete SR pipeline should rely on the present results.

What is at stake. Our meta-analytic propagation analysis reveals that 5–21% of articles produce different numbers of effect estimates across runs, meaning the very composition of a meta-analysis depends on which LLM run generated its inputs. As LLMs become embedded in systematic reviews informing clinical guidelines and public health policy, this hidden variability cannot remain unexamined.

The path forward is clear: multiple repetitions, provenance hashing, and transparent variation reporting must become standard practice in LLM-assisted evidence synthesis. Our open-source infrastructure, complete dataset, and provenance protocol provide the tools to make this transition immediately actionable.

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Use of AI Writing Tools. AI-based tools (Claude, Anthropic) were used for code development assistance and manuscript editing support. All scientific content, experimental design, data analysis, and interpretation of results were performed exclusively by the human authors. The AI tools used in manuscript preparation are the same models evaluated as experimental subjects in this study; this dual role was managed by ensuring that all reported results were computed deterministically from the raw experimental data, independently of any AI writing assistance. **Timeline disclosure for evaluator-as-writing-aid concern.** The reproducibility results (`analysis/reproducibility_results.json`) were committed to the project repository on **2026-03-02** (commit `1b90c1b`), after deterministic computation from raw experimental outputs (`data/raw_outputs/`, committed 2026-02-13 through 2026-03-02 across all six models). The first complete manuscript draft was committed on **2026-03-11** (commit `646c6a3`); the post-revision blindage analyses were committed on **2026-04-23 to 2026-04-28**. Claude Sonnet 4.5—one of the evaluated models—was used as a writing aid during manuscript preparation only after results were locked at the 2026-03-02 commit. All numerical results, tables, and figures derive deterministically from the locked output hashes; no result depends on the writing-time use of any LLM. The complete commit history is publicly auditable at https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility.

Data Availability Statement. All code, data, prompts, schemas, raw outputs, and run provenance records are publicly available at https://github.com/Roverlucas/llm-evidence-synthesis-reproducibility and archived at the Open Science Framework (https://doi.org/10.17605/OSF.IO/VR934). The dual-human labeling protocol is deposited separately as a frozen OSF registration (https://doi.org/10.17605/OSF.IO/FGN3E, 2026-05-12), which pre-dates label collection; the returned labels, agreement statistics, and the v1.2 protocol amendment are in the repository under `data/dual_labeling/` (**[PENDING: repository tag for the submitted version]**). API keys are excluded from the repository. The 500 PubMed abstract identifiers, gold standard labels, and complete analysis scripts are included to enable full reproduction of the reported results.

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